Apparatus

Paul Gordan, *Ueber die Transformation der Thetafunctionen* (1863)

Final repaired critical-diplomatic edition

1. Editorial status and authority

This apparatus accompanies two independent products: a German diplomatic transcription and a faithful English rendering. It is not a corrected mathematical edition. The German body preserves the printed wording, spelling, punctuation, and mathematical anomalies wherever the pixels are legible. The English body translates the prose while retaining the same mathematical readings. Clarifications and rejected inherited readings are confined to the numbered notes below.

The controlling witness is the 20-unit authority PDF with SHA-256

`c9b744640aa68f448f432bd495c68129a108dd15f6480fdb725aad1b035f73aa`

The convenience page-image witnesses were checked against that PDF. The edition uses stable invisible destinations `source-PA01` through `source-PA20`; the TeX source exposes them as `\sourceanchor{PAxx}`.

2. Exact source topology and disposition

Unit	Witness role	Scholarly disposition
PA01	Title page	Authorial title matter transcribed and translated.
PA02	Blank verso with copy-specific marks	Library stamp, identifier, and digitization matter excluded from the scholarly text; exclusion recorded.
PA03	Printed text: §§1–3 opening	Transcribed; relations 1–6 and the statement of §3 included.
PA04	§3 proof; §4 opening	Transcribed from the first "Beweis" through the opening of the second proof.
PA05	§4 continuation	Transcribed through the double-series evaluation.
PA06	§§5–6	Transcribed through the differential-equation computation.
PA07	§6 conclusion; §§7–9	Transcribed through the first paragraph of §9.
PA08	§9 continuation; §10	Transcribed through the iterative transformation system.
PA09	§10 continuation; §§11–12 opening	Transcribed through the first PDE substitution in §12.
PA10	§12 continuation	Transcribed through equation 10 and the lead-in to formula 11.
PA11	Formula 11; §§13–14	Transcribed through the substituted relation at the foot of the page.
PA12	§14 continuation; §§15–16	Transcribed through formula 17.
PA13	§17 opening	Transcribed through the first finite Gaussian sum.
PA14	§17 continuation; formula 18; §18 opening	Transcribed; odd-γ case completed.
PA15	§18 continuation	Transcribed; even-γ case and the 2γ-term summation continued.
PA16	§18 conclusion; §§19–20 opening	Transcribed through the lead-in to the system used in §20.
PA17	§20 continuation; §21	Transcribed through the final identity of §21.
PA18	Alternate final-page witness	Corroborates §22 and the ornament; authorial content not duplicated.
PA19	Blank unit	No authorial content; recorded as blank and not transcribed.
PA20	Controlling full final page	Supplies the controlling reading of §22 and the printer's ornament.

3. Numbered critical and technical notes

GTH-A01 (PA02, exclusion). The visible stamp, shelf or accession identifiers, and Google digitization watermark belong to the scanned copy and its reproduction history, not to Gordan's printed work. They are retained in the authority files and exclusion ledger but omitted from all three scholarly products.

GTH-A02 (PA03, §2, relations 4–5). The page pixels include i after π in the modulus shifts: $a + 2n\pi i$ and $a + n\pi i$. Earlier inherited text without these two i's was rejected.

GTH-A03 (PA04, §3). The first displayed Fourier expansion prints $e^{x_\alpha+\alpha ib}$, while the subsequent expansions print $e^{x_\alpha+2ib\alpha}$. The German transcription preserves this discrepancy; the English rendering does the same. No silent factor-of-two emendation has been made.

GTH-A04 (PA10, §12). The source reads

$$\varphi(a) = \frac{i\pi}{\gamma a + \delta i\pi},$$

with a positive numerator. A negative inherited variant was rejected after direct pixel inspection.

GTH-A05 (PA11, formula 11). The printed first argument of the final Θ-function is

$$\frac{\alpha a + \beta i\pi}{\gamma a + \delta i\pi},$$

without a leading $i\pi$. This is inconsistent with the definition $A = i\pi(\alpha a + \beta i\pi)/(\gamma a + \delta i\pi)$ on PA09 and with later formulae. Because the owned product is diplomatic, the omission is preserved in both bodies. The mathematically motivated inherited insertion of $i\pi$ is reported here, not silently adopted.

GTH-A06 (PA12, §16). In the transformed argument and in formula 17 the pixels read $\mu\omega\pi/\gamma$. The tempting inherited replacement $\mu\alpha\pi/\gamma$ was rejected. This note records the source-level reading without attempting normalization.

GTH-A07 (PA14, §17). In the odd-γ derivation the second Θ-argument contains the printed term $-\pi\gamma\delta(1+\lambda+\beta)/2$. The factor π is present and is retained.

GTH-A08 (PA15, §18 opening). The source is asymmetric: the left-hand argument ends with $\mu\pi/\gamma$, whereas the right-hand argument ends with $\alpha\mu\pi/\gamma$. The inherited form with $\alpha\mu\pi/\gamma$ on both sides was rejected.

GTH-A09 (PA16, settings before formula 19). The middle equality is printed $i\rho/2$, not $i\omega/2$:

$$b\sqrt{\frac{\omega}{\pi}} = \frac{i\rho}{2} = -\frac{i\pi}{2\gamma}.$$

The source reading is retained even though ρ is not otherwise introduced in the immediate passage.

GTH-A10 (PA16, formula 19). The exponential factor includes the leading γ: $e^{\gamma i\pi(2\beta+2-\delta)/4}$. The printed formula does not include an additional $e^{\gamma b^2/(\gamma a+\delta i\pi)}$ factor; none has been supplied editorially.

GTH-A11 (PA18–PA20, final-page topology). PA18 and PA20 witness the same final printed page. PA18 is treated as the alternate witness specified by the packet; PA19 is blank; PA20 controls the final body text and ornament. The authorial text is transcribed once only.

GTH-A12 (PA01, title-page ornaments). The source contains a short divider beneath the title and a larger ornamental rule above the imprint. Both are preserved by lossless raw crops plus conservative presentation derivatives rather than generic TeX rules. The source image is 575 by 737 pixels. The half-open crop boxes, raw dimensions, derivative dimensions, and hashes are:

Asset	Crop box; dimensions	Operation	SHA-256
PA01 divider raw	(215,230,312,243); 97x13	none	c113b88cccd47533801e05328bce99881201390b5731c5892ad3564aa1612bc6
PA01 divider derivative	source as above; 388x52	grayscale; full-range autocontrast; 4x Lanczos	120eefa8c365d13205e64280a42239dcbf8003dc789abb272d4dc24adfd17c4a
PA01 ornament raw	(110,578,420,598); 310x20	none	54f702f6cc52bbde1ec9a875efa9e320a58ae47fc62974fe38c8b88910fb8070
PA01 ornament derivative	source as above; 1240x80	grayscale; full-range autocontrast; 4x Lanczos	5ad14efd1b32d70399e6ed27a09cadf4baaad456f52510bd13755ea3d5b6b7eb



GTH-A13 (PA20, final printer's ornament). The final ornament is not approximated with generic TeX glyphs. A raw crop and a conservative presentation derivative are packaged. The raw source image is 575 by 741 pixels. The half-open crop box is $(x_0, y_0, x_1, y_1) = (245, 360, 375, 395)$, yielding 130 by 35 pixels. The derivative applies grayscale preservation, full-range autocontrast, and 4x Lanczos resampling only, yielding 520 by 140 pixels. Hashes are:

```
raw:  664de1d2550bcccbb5db7117fcf091409c14b80958e1b9bacaf605bb5a5ce38e
derivative:  b2969b2c39fb08c626b8683ec8a70203d0165ed30be85906f665e5a374d13fba
```



GTH-A14 (PA08, §10, isolated definition of a_2). The isolated definition prints

$$a_2 = -\frac{\pi^2}{a_1 + 2\lambda_1 i\pi},$$

with a minus sign. The full iterative system later on the same page prints the corresponding definition with a positive numerator. Both local readings are retained in the diplomatic German and English bodies. The later positive form may indicate that the isolated minus is a typographical error, but no silent emendation is made.

GTH-A15 (PA08, §10, isolated second transformation equation). In the equation immediately following the isolated definition of a_2, b_2, the printed left-hand side is $\Theta(a, b)$, not $\Theta(a_1, b_1)$. The subsequent full iterative system does print $\Theta(a_1, b_1)$ at the corresponding stage. The diplomatic bodies preserve this local inconsistency; $\Theta(a_1, b_1)$ is the contextually expected mathematical reading.

GTH-A16 (PA11, §14, parameterization of δ). The final equality is printed with vertical delimiters:

$$\delta = \delta_1 + \gamma_1\delta_1(\alpha_1 - \beta_1) = \alpha_1\delta_1|\gamma_1 + \delta_1|.$$

The preceding β-line visibly uses braces, so the delimiter distinction is not normalized. The bars are reproduced diplomatically without imposing an interpretation beyond the printed glyphs.

GTH-A17 (PA14, formula 18). Formula 18 prints the prefactor denominator $\varphi(2\alpha, \beta)$. The derivation immediately above ends with $\varphi(2\alpha, \gamma)$, so γ is the mathematically motivated correction. The diplomatic bodies retain the printed β; the probable correction is recorded only here.

GTH-A18 (PA16, derived value of c before formula 19). The source line reads

$$2\sqrt{\gamma}\, e^{\gamma i\pi(2\beta+2-\delta)/4} = c\varphi(\alpha, 2\gamma),$$

with no minus sign in the exponent. Formula 19 immediately below also carries the positive exponent. Both monolingual bodies now preserve that reading.

4. Formula, figure, and layout policy

All mathematical expressions are encoded as editable TeX. There are no authorial tables or diagrams. The non-formula graphical elements are the two title-page ornaments and the final printer's ornament, handled under GTH-A12–A13. The five repaired source-fidelity readings are documented under GTH-A14–A18. Lineation and page makeup are not reproduced as a facsimile; paragraph order, section sequence, equation labels, formula membership, and the relative order of all visible authorial elements are preserved. Page-unit anchors permit audit back to each physical witness.

5. Build and QA statement

The three PDFs are built with pdfTeX under a fixed `SOURCE_DATE_EPOCH`, omitted creation dates, and fixed trailer identifiers. Clean rebuilds are byte-identical. Automated checks cover TeX failure, box warnings, PDF page counts, text extraction, source-anchor presence, German/English display-math parity, ZIP integrity, and SHA-256 manifests. Rendered contact sheets were visually inspected for clipping, overlap, broken glyphs, missing equations, and ornament placement. The machine-readable audit files in `05_audit/` carry the detailed results.

6. Bibliographic identification

Gordan, P. (1863). *Ueber die Transformation der Thetafunctionen.* Giessen: Druck von Wilhelm Keller.